Leopold **Mareis**

Statistician · Data Scientist · Quantitative Analyst

Munich, Germany

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Profile

Statistician and data scientist finishing a Ph.D. in mathematical statistics at TU Munich, specialising in causal inference for reliable, uncertainty-aware decision-making. Five years of applied experience across reinsurance, clinical trials and industrial R&D, developing statistical and machine-learning solutions in R, Python and Julia and communicating results to boards and non-technical stakeholders. Available from December 2026 and based in Munich.

Experience

Technical University of Munich (MaRDI / NFDI)
Munich, Germany

Scientific Researcher · Research Data & Causal Inference — *Dec. 2024 – Present*

- Conducted statistical research on causal inference and experimental design as lead author, both independently and in collaboration; published and presented at leading venues (CLeaR, PGM, IMS).
- Sole architect and developer of an open research-data platform for graphical modelling and causal inference (NFDI-funded MaRDI initiative), grown to **14k views and 8k downloads**.
- Curated benchmark datasets and wrote hands-on notebooks that make causal-inference methods reproducible and reusable for teaching and outreach.
- Developed graphical-modelling tooling in Julia (OSCAR.jl) and presented it to the open-source community at JuliaCon 2026.

Fraunhofer Institute for Cognitive Systems (IKS)
Munich, Germany

Scientific Researcher · Reasoned AI Decisions — *Oct. 2021 – Nov. 2024*

- Led the causal analysis of the **BIOTRONIK BIOSOLVE-II** cardiovascular-stent trial (paid industry study), presenting effect estimates with quantified uncertainty to the client's development board and recommending concrete procedural changes to reduce target-lesion failures.
- Prototyped counterfactual reinforcement-learning decision support for the **ADVITOS ADVOS multi** ICU multi-organ-support device, informed by dedicated experiments the partner ran exclusively for this work.
- Built and open-sourced causal-inference libraries in R and Python (PARCS: Partially Randomized Causal Simulator), documented in an arXiv preprint and reused across projects.
- Advised on statistical methodology for the development and regulatory certification of clinical AI software.

Munich Re
Munich, Germany

Intern · Corporate Underwriting (Exposure & Loss Modelling) — *2019*

- Modelled loss-severity distributions and loss-relief curves in R for exposure analysis.
- Built and automated SAS reporting pipelines (Enterprise Guide, Visual Analytics) on a large property-loss database supporting underwriting.
- Cleaned and consolidated a large claims database to improve data quality for analysis.
- Reviewed facultative reinsurance submissions alongside underwriters.

Agrilution, indoor-farming startup
Munich, Germany

Intern — *2018*

Skills

Programming	R, Python, Julia, Git, Docker, LaTeX, Quarto, Vim, command-line workflows, AI-assisted development (Claude Code)
Statistics & ML	Causal inference, semiparametric graphical modelling, GLMs & regression, copulas & quantile regression, experimental design, reinforcement learning, uncertainty quantification
Domains	Financial & insurance risk modelling, clinical trials & biostatistics, industrial R&D
Languages	German (native), English (fluent), French (basic)

Education

Technical University of Munich *Munich, Germany*

Ph.D. in Mathematical Statistics (Dr. rer. nat.) *Dec. 2021 – Dec. 2026 (expected)*

- Thesis: *Asymptotic Inference in Linear Causal Models*; advised by Prof. Dr. Mathias Drton.

Technical University of Munich *Munich, Germany*

M.Sc. in Mathematics in Operations Research *2018 – 2021*

- Thesis on vine-copula-based quantile regression with discrete components (Prof. Dr. Claudia Czado), applied to dependence modelling for risk and finance.
- Semester abroad at TU Delft, Netherlands.

Technical University of Munich *Munich, Germany*

B.Sc. in Mathematics *2015 – 2018*

Selected Publications

2026	**CLeaR**, Cost-Aware Optimized Front-Door Experimental Design. **L. Mareis**, M. Drton
2026	**PGM**, Semiparametric Inference for Half-Trek Estimators in Linear SEMs. **L. Mareis**, N. Sturma, M. Drton
2025	**Separation & Purification Technology**, Low-Gradient Magnetic Separation in Biotechnology: Parameter Identification and Model-Guided Scale-Up. Meyer et al., incl. **L. Mareis**
2026	**Under review**, Causes and Effectiveness of Gender-Equality Instruments in Organizations. I. Hagerer, **L. Mareis**, D. Schkoda

Full list of publications and talks at lmareis.github.io.

Selected Talks

2026	**IMS Annual Meeting**, Cost-Aware Optimized Front-Door Experimental Design
2026	**JuliaCon**, Graphical Modelling with Symbolic Algebra in OSCAR.jl
2024	**bitkom AI & Data Summit**, Counterfactual Decision Support for a Clinical Multi-Organ ICU Machine
2023	**KI-Medtech Congress**, Estimating Causal Effects of Stent Operations

Awards, Teaching & Supervision

2026	**Student Travel Grant**, Conference on Probabilistic Graphical Models (PGM)

Supervision: advised 4 M.Sc. theses and 2 university seminars at TU Munich in graphical modelling, causal inference and machine learning, with applications in biology, finance and autonomous driving.

Teaching: teaching assistant for 5 courses, including Computational Statistics, Graphical Models in Statistics, and Statistics for Business Administration.

Leadership & Interests

State-certified choir conductor and, since 2024, elected youth representative on the board of the Bavarian Choral Association (BSB), leading rehearsals, organising events and coordinating volunteer teams. Passionate about classical music and choral singing.